
End-to-End Staff-to-Jianpu Translation with a Multi-Head Vision-Encoder-Decoder

BEN-LU CHEN CHUN-JUI HSU MENG-XI LIN CHIEN-AN CHOU

Institute of Electrical and Computer Engineering

National Yang Ming Chiao Tung University

{sino.ee14, ray30247905.ee10, s111511265.ee11, cityjd.ee11}@nycu.edu.tw

Abstract

We present an end-to-end deep-learning system that translates images of printed monophonic staff notation directly into numbered musical notation (Jianpu), bypassing the intermediate MusicXML/MIDI formats and hand-written grouping heuristics of conventional Optical Music Recognition (OMR) pipelines. A Vision-Encoder-Decoder ingests a score image and a Transformer decoder emits four *decoupled* token streams—type, pitch, rhythm, and attribute—so that pitch and rhythm accuracy can be measured independently; a post-processor maps the streams to Jianpu, and the whole pipeline is shipped as an interactive demo. We train and evaluate exclusively on a fully synthetic corpus built with *music21* and *verovio*, whose generator emits the four label streams as ground truth in lock-step with the rendered image. Three findings emerge. (i) An ImageNet-pretrained ViT encoder with an autoregressive (AR) decoder solves the task almost perfectly (symbol error rate, SER, 0.0029), a from-scratch ResNet encoder under the same AR decoder *structurally fails to learn pitch* ($\approx 0\%$), yet attaching a BiLSTM + per-head CTC decoder to the *same* ResNet features recovers a perfect SER of 0.0000—showing the bottleneck is the AR monotonic alignment over 1-D column features, not the convolutional encoder. (ii) In a sample-efficiency sweep, CRNN+CTC reaches usable quality from 5,000 samples while the ViT-AR pitch head does not begin to learn until 50,000, contradicting the expectation that ImageNet pretraining helps in the low-data regime. (iii) A photo-augmentation fine-tune more than halves SER on real phone photographs of scores ($0.634 \rightarrow 0.258$), and a leave-one-out ablation isolates geometric (camera-angle) augmentation as the dominant factor.

1 Introduction

Converting standard five-line staff notation into numbered musical notation (Jianpu) is a tedious yet essential task. In traditional Chinese music clubs and ensembles, Jianpu is the standard for sight-reading and performance, and an automated tool that directly outputs readable Jianpu would eliminate hours of manual transcription, letting musicians focus on playing rather than on score translation. Despite this practical demand, the problem is largely open: current Optical Music Recognition (OMR) research overwhelmingly targets the digitisation of *Western* notation into intermediate formats such as MusicXML or MIDI. Seminal pipelines [Calvo-Zaragoza et al., 2020] and recent open-source systems [Yoyo et al., 2023] rely on multi-stage semantic segmentation followed by heuristic note-grouping rules; these stages accumulate cascading errors and are not designed for cross-format (staff-to-Jianpu) translation.

We instead treat the task as a single image-to-sequence translation problem. A Vision-Encoder-Decoder ingests a 2-D image of a printed monophonic score and autoregressively decodes a 1-D sequence of custom semantic tokens, which a post-processor maps to Jianpu. By avoiding intermediate representations and hand-coded heuristics, the network must implicitly learn the music-theoretic

relationship between the 2-D spatial arrangement of symbols and the 1-D target. Three challenges shape the design:

- **Challenge 1: the 2D-to-1D semantic gap.** Pitch is encoded by the *absolute vertical position* of a notehead on the staff, while rhythm and ordering are largely horizontal. The model must reconcile both axes into one ordered token stream.
- **Challenge 2: translation without intermediate representations.** Bypassing MusicXML/MIDI means the model learns alignment and notation conventions purely from visual supervision, with no explicit rule engine to fall back on.
- **Challenge 3: data scarcity.** No large paired staff-to-Jianpu corpus exists, so training must rely on synthetic data while still generalising to real, photographed scores.

To make pitch and rhythm *separately* measurable—and to expose where a model fails—we decouple the target into four parallel streams (*type, pitch, rhythm, attribute*) emitted in lock-step by the generator, and give the decoder four corresponding output heads. This design turns the architecture comparison into a controlled experiment: holding the decoder, loss, and data fixed, we can attribute a pitch failure to the visual representation rather than to the language model. Our main contributions are:

1. **An end-to-end OMR→Jianpu system.** A fully synthetic data pipeline (music21+verovio) that emits four decoupled label streams as ground truth, a multi-head Vision-Encoder-Decoder, a token-to-Jianpu post-processor, and a deployed interactive demo.
2. **An architecture finding that reframes a “ResNet cannot learn pitch” result.** Under an autoregressive (AR) decoder, a pretrained ViT encoder solves the task (SER 0.0029) while a from-scratch ResNet encoder structurally fails on pitch ($\approx 0\%$). Attaching a BiLSTM + per-head CTC decoder to the *same* ResNet features reaches a perfect SER of 0.0000, showing the true bottleneck is the AR monotonic alignment over 1-D column features—not the convolutional encoder.
3. **A sample-efficiency study.** CRNN+CTC reaches usable transcription from 5,000 samples whereas the ViT-AR pitch head does not begin to learn until 50,000, contradicting the common expectation that ImageNet pretraining helps most in the low-data regime.
4. **A sim-to-real study.** A photo-augmentation fine-tune more than halves SER on real phone photographs ($0.634 \rightarrow 0.258$), and a leave-one-out ablation isolates geometric (camera-angle) augmentation as the dominant factor.

This is the Application track (NYCU 535354) final report; the code, synthetic generator, trained checkpoints, and demo accompany the submission.

2 Related Work

Pipeline OMR. Classical OMR decomposes recognition into staff-line detection, symbol segmentation/classification, and rule-based reconstruction of musical semantics; Calvo-Zaragoza et al. [2020] survey this paradigm and its failure modes. Modern open-source systems such as Oemer [Yoyo et al., 2023] replace hand-crafted segmentation with U-Net semantic segmentation but still rely on downstream heuristics to group primitives into notes and to recover voice/rhythm structure. These multi-stage designs accumulate cascading errors and emit Western intermediate formats (MusicXML/MIDI); converting their output to Jianpu would require yet another heuristic stage. We avoid the pipeline entirely and decode the target sequence directly.

End-to-end image-to-sequence recognition. Treating recognition as sequence transduction has a long history in OCR and scene-text reading. The CRNN [Shi et al., 2016] couples a convolutional feature extractor with a recurrent network trained under the Connectionist Temporal Classification (CTC) loss [Graves et al., 2006], which marginalises over all monotonic alignments between a feature sequence and a shorter label sequence and so needs no per-frame supervision. For music specifically, Calvo-Zaragoza and Rizo [2018] show that a CRNN+CTC model transcribes *monophonic* scores end-to-end, motivating CTC as a natural baseline for our setting. More recently, Transformer Vision-Encoder-Decoders such as TrOCR [Li et al., 2023] pair a pretrained vision encoder [Dosovitskiy et al.,

2021] with a pretrained autoregressive text decoder [Lewis et al., 2020], reaching state-of-the-art OCR via cross-attention rather than CTC’s monotonic alignment. Our architecture follows this Vision-Encoder-Decoder template but replaces the single text vocabulary with four decoupled music-semantic heads, and we directly contrast the AR decoder against a CTC decoder on identical convolutional features—a comparison that, to our knowledge, has not been reported for the staff-to-Jianpu task.

Synthetic data for music notation. Because paired staff-to-Jianpu corpora do not exist, we generate data programmatically. We build symbolic scores with `music21` [Cuthbert and Ariza, 2010] and engrave them to images with `verovio` [Pugin et al., 2014]; crucially, the generator emits the ground-truth token streams *itself*, so labels are exact rather than parsed back from the rendered image. This lets us study sample efficiency and sim-to-real transfer under fully controlled supervision.

3 Method

We frame staff-to-Jianpu transcription as supervised image-to-sequence translation. The model receives a 2-D image of a printed monophonic score and decodes a 1-D sequence of semantic tokens; a deterministic post-processor then renders those tokens as Jianpu. Training supervision comes entirely from a synthetic corpus whose generator emits the target tokens directly, so labels are exact. We first describe the data and its four-stream label contract (§3.1), then the shared encoder-decoder model (§3.2), the two decoder variants we compare (§3.3), the objective and optimisation (§3.4), and finally the Jianpu mapping and evaluation metrics (§3.5).

3.1 Synthetic data and the four-stream label contract

Each sample is built by `music21` [Cuthbert and Ariza, 2010] as a symbolic `Stream`, exported to `MusicXML`, and engraved to a PNG by `verovio` [Pugin et al., 2014] (pinned to a fixed version, since glyph spacing is version-sensitive). Generation is fully deterministic: a sample’s RNG is derived from `SeedSequence([seed, sample_idx])`, so the pair $(seed, index)$ reproduces the same score and labels across runs. The scope is deliberately restricted to *monophonic* printed notation—no chords, slurs, or handwriting—so that the target is an unambiguous single-voice sequence.

Crucially, the generator emits the ground-truth labels itself, in lock-step with each appended symbol, as **four parallel decoupled streams**:

- **type** $\in \{\text{note, rest, barline, clef, key_signature, time_signature}\}$;
- **pitch** (notes only): letter $\times$ accidental (bb/b/♭/♯/##) $\times$ octave 2–6, otherwise `<NULL>`;
- **rhythm** (notes/rests only): whole–32nd, optionally dotted, otherwise `<NULL>`;
- **attribute** (clef/key/time only): clef (G2/F4/C3/C4), key signature in fifths (ks-7..ks+7), or time signature, otherwise `<NULL>`.

The four streams are row-aligned (one entry per emitted symbol), so pitch and rhythm can be scored independently (Figure 1).

The closed vocabularies have 6, 175, 12, and 26 content tokens respectively, plus shared structural specials (PAD=0, BOS=1, EOS=2, UNK=3, NULL=4) frozen across all four vocabularies so a single embedding index denotes a structural position in every stream. Label rows carry no leading BOS: they are `[content . . . , EOS, PAD . . .]`, and the decoder shifts in BOS itself (§3.2).

3.2 Multi-head Vision-Encoder-Decoder

A visual encoder maps the score image to a sequence of feature tokens, and a Transformer decoder cross-attends to them to emit the four streams (Figure 2). We compare two encoders under an identical decoder. The `vit` encoder is an ImageNet-pretrained Vision Transformer [Dosovitskiy et al., 2021] with the `[CLS]` token dropped, fed a $3 \times 224 \times 224$ aspect-preserving, white-padded image and producing 196 feature tokens. The `resnet` encoder is a from-scratch convolutional stack (five strided blocks, channels $32 \rightarrow 256$) over a $1 \times 128 \times W$ grayscale image; it pools the staff height into one token per horizontal column, yielding a variable-length feature sequence with a padding mask.

The decoder is a BART-style [Lewis et al., 2020] Transformer ($d_{\text{model}} = 384$, 4 layers, 6 heads, feed-forward dimension 1536, dropout 0.1). Rather than one output vocabulary it carries **four**

3.4 Training objective and optimisation

The objective is the sum of per-head cross-entropy losses over the four streams. PAD positions are ignored. Two optional knobs address class imbalance: (i) a *NULL mask* that, for the pitch/rhythm/attribute heads, coerces NULL targets to the ignore index so the head is not dominated by the many positions where it is structurally inapplicable; and (ii) an *EOS up-weighting* on the type head, since the single EOS per sequence is otherwise swamped by note/rest tokens and the model under-learns when to stop.

All runs use AdamW (learning rate 5×10^{-4} , weight decay 0.01) with a cosine schedule (1,000 warm-up steps), 30 epochs, bf16 mixed precision, and gradient clipping at 1.0, orchestrated with Hydra and accelerate. We train on 100,000 deterministic synthetic samples (generated on the fly) and select the checkpoint with the best validation SER on a held-out, pre-rendered set of 1,000 samples; a separate 1,000-sample test set is rendered once for final reporting.

3.5 From tokens to Jianpu, and evaluation metrics

Post-processing is type-anchored: the type stream is the structural authority, and a small state machine tracks the running clef, key signature, and time signature as those tokens appear, mapping each note’s (letter, accidental, octave) to a Jianpu scale degree, chromatic alteration, and octave-dot count via a circle-of-fifths tonic lookup. The mapping is deterministic and total—a wild prediction outside the valid key range yields an empty Jianpu string rather than a crash—so it is safe to run on free-form model output at deploy time.

We report three metrics, all computed by aligning the predicted and ground-truth sequences as composite 4-tuples (*type, pitch, rhythm, attribute*) with Levenshtein edit operations (substitution counts iff any field differs):

- **Symbol Error Rate (SER)**, our primary metric: edit distance divided by ground-truth length, analogous to character error rate;
- **Pitch accuracy**: among aligned ground-truth notes, the fraction with the correct pitch token;
- **Rhythm accuracy**: among aligned ground-truth notes/rests, the fraction with the correct rhythm token.

Decoupling pitch and rhythm from the overall SER is what lets us localise the encoder failure in §4.

4 Experiments

All experiments use the synthetic corpus of §3.1: training samples are generated on the fly, and we evaluate on a held-out, pre-rendered 1,000-sample validation set (used for checkpoint selection) and an analogous 1,000-sample test set. Unless noted, the decoder, loss, optimiser, and 30-epoch budget are held fixed at the values in §3.4; only the factor under study varies. We organise the study around three questions: which visual representation and decoder actually solve the task (§4.1), how the two paradigms scale with data (§4.2), and whether the model transfers to real photographs (§4.3). §4.4 describes the deployed demo.

4.1 Which representation learns pitch?

We hold the decoder, loss, and data fixed and vary only the visual backbone and the decoding paradigm, at 100,000 training samples. Table 1 reports the result and Figure 3 the corresponding autoregressive validation curves.

Findings. Under the AR decoder, the pretrained vit encoder solves the task almost perfectly ($SER = 0.0029$, 99.85% pitch), while the from-scratch resnet encoder learns rhythm only partially and **structurally fails to learn pitch** ($\approx 0\%$): its pitch cross-entropy is pinned at the “NULL-only” floor, i.e. it is correct on non-note positions but uninformative on notes. A translation-equivariant CNN that pools image height into one token per column discards the *absolute vertical position* of a notehead—which is exactly what pitch is on a staff. This failure proved invariant to learning-rate sweeps, to replacing height mean-pooling with a height-flatten, to higher vertical resolution, and to

Table 1: Architecture comparison on the 1,000-sample held-out synthetic validation set (100,000 training samples, 30 epochs). Same data and four-head targets throughout; only the encoder and decoder differ. AR = autoregressive decoder; CTC = BiLSTM + per-head CTC decoder.

Encoder + decoder	Val SER ↓	Pitch Acc. ↑	Rhythm Acc. ↑
vit (ImageNet-pretrained) + AR	0.0029	99.85%	99.78%
resnet (from-scratch CNN) + AR	~1.01	~0%	~40%
resnet + BiLSTM + CTC (CRNN)	0.0000	100%	100%

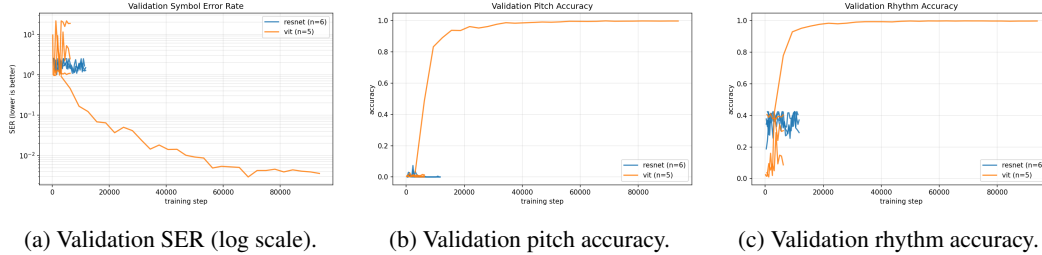


Figure 3: Autoregressive validation metrics over training for the two encoders (*vit* vs *resnet*). The *vit* backbone drives SER to ≈ 0.003 and pitch/rhythm to $\approx 100\%$, whereas *resnet* stalls at SER ≈ 1 and never learns pitch.

an explicit learned vertical positional embedding, which is why the natural conclusion was that the convolutional encoder simply cannot represent pitch.

The CTC row refutes that conclusion. Feeding the *same* column-pooled ResNet features to a BiLSTM + per-head CTC decoder reaches a *perfect* SER = 0.0000 with 100% pitch and rhythm. The encoder was never the bottleneck; the AR failure was a joint property of monotonic, one-column-per-symbol cross-attention over 1-D features. CTC’s flexible alignment plus the BiLSTM’s bidirectional context let each column read the staff-line geometry of its neighbours as a vertical reference frame, recovering pitch that AR could not.

4.2 Sample efficiency

To probe how the two paradigms use data, we sweep both across five training-set sizes (1,000 to 100,000), holding everything else fixed (Table 2, Figure 4).

Table 2: Best validation SER, pitch, and rhythm accuracy reached within 30 epochs at each training-set size, on the 1,000-sample held-out synthetic split. Single seed per cell.

Model	Train size	Best val SER ↓	Pitch Acc. ↑	Rhythm Acc. ↑
CRNN+CTC	1,000	0.9484	0.00%	0.77%
CRNN+CTC	5,000	0.0671	97.14%	97.28%
CRNN+CTC	20,000	0.0100	99.95%	99.61%
CRNN+CTC	50,000	0.0002	99.98%	100.00%
CRNN+CTC	100,000	0.0000	100.00%	100.00%
ViT-AR	1,000	1.0714	1.07%	28.08%
ViT-AR	5,000	0.8658	3.68%	32.46%
ViT-AR	20,000	0.8938	0.24%	28.52%
ViT-AR	50,000	0.0318	97.48%	98.20%
ViT-AR	100,000	0.0029	99.85%	99.78%

Findings. The two paradigms occupy qualitatively different sample-efficiency regimes. CRNN+CTC reaches usable transcription already at 5,000 samples (val SER 0.067, pitch 97.1%) and is within 10^{-3} of perfect SER from 50,000 upward. ViT-AR instead exhibits a sharp data threshold: at $\leq 20,000$ samples its pitch head fails to learn at all (pitch $\leq 3.7\%$, val SER stuck near 0.9, the

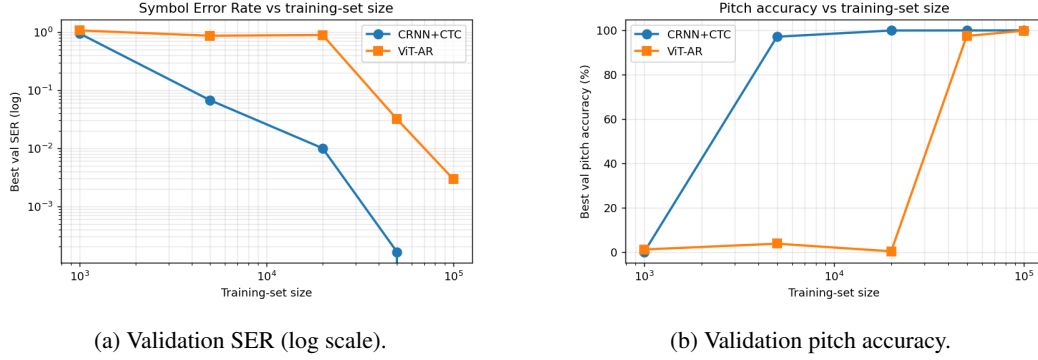


Figure 4: Sample efficiency vs. training-set size for CRNN+CTC and ViT-AR. CRNN’s curves drop between 1k and 5k samples; ViT’s drop only between 20k and 50k.

regime where free-running generation degenerates into runaway sequences), then snaps to val SER 0.0318 / pitch 97.5% at 50,000, close to its 100,000-sample asymptote.

This contradicts the common expectation that ImageNet-pretrained features dominate the low-data regime. The pretrained ViT prior carries no music-notation structure: pitch depends on a notehead’s absolute vertical position on the staff, a cue ImageNet pretraining does not reward. Combined with AR’s one-column-per-symbol alignment, the pitch head needs far more data before it escapes the NULL-only floor, whereas CRNN+CTC’s bidirectional context and flexible alignment make per-symbol learning proportionately more sample-efficient. (We report a single seed per cell; the ViT-AR threshold may shift with seed and data loader settings, but the order-of-magnitude gap is robust—see §5.)

4.3 Sim-to-real transfer

The deterministic generator lets us photograph the *same* val/test samples with a phone and score predictions quantitatively, since each photo pairs 1:1 by filename with its synthetic ground-truth labels. The clean-trained vit model collapses on these photos (Table 3, top): test SER jumps from 0.003 (clean) to 0.634 and pitch from 99.8% to 32.5%. We then warm-start from that checkpoint and fine-tune for 3 epochs (LR 5×10^{-5}) under a photo augmentation profile that adds, in order, native-resolution lighting/degradation (brightness, gamma, shadows, blur, noise, JPEG) and, after padding to the 224^2 canvas, geometric warps (small rotation, shear, perspective). Fine-tuning more than halves the error (test SER 0.258, pitch 64.5%) while clean validation barely regresses.

Table 3: Real photographed scores: clean-trained baseline vs. photo fine-tune. n is the number of photos; *val/test* are the external photo splits (paired to the synthetic val/test samples), *all* their union.

Model	Split	n	SER ↓	Pitch Acc. ↑	Rhythm Acc. ↑
baseline	val	100	0.5858	0.3353	0.8778
baseline	test	100	0.6343	0.3248	0.8617
baseline	all	200	0.6097	0.3302	0.8699
fine-tuned	val	100	0.2409	0.6686	0.9781
fine-tuned	test	100	0.2582	0.6454	0.9648
fine-tuned	all	200	0.2494	0.6574	0.9716

What in the augmentation matters? A leave-one-out ablation (Table 4) drops each augmentation group in turn; each variant is selected on the external *val* split and reported on external *test*. Removing the geometric group alone reverts photo performance almost to baseline (SER 0.605, pitch 0.331), a swing of +0.347 SER / −0.314 pitch—far larger than dropping lighting (+0.068 / −0.082) or degradation (+0.015 / −0.011). A control that fine-tunes with the default (non-photo) profile matches the baseline on photos, confirming the gain comes from the augmentation, not the extra training steps. The geometric group’s dominance pinpoints the sim-to-real gap as camera-angle skew

causing vertical mis-registration of noteheads—consistent with the residual pitch errors being mostly “same octave, wrong step.”

Table 4: Augmentation leave-one-out on the photo fine-tune (warm-started from the clean vit model; 3 epochs each). Selected on external *val*, reported on external *test*; clean columns are the synthetic validation set.

Condition	Aug profile	Test SER ↓	Test pitch ↑	Clean SER ↓	Clean pitch ↑
baseline (no fine-tune)	—	0.634	0.325	0.0030	0.9983
fine-tune, no photo aug	default	0.634	0.314	0.0016	0.9990
photo, full	photo	0.258	0.645	0.0083	0.9917
photo — geometric	photo_no_geom	0.605	0.331	0.0034	0.9980
photo — lighting	photo_no_light	0.326	0.563	0.0117	0.9859
photo — degrade	photo_no_degrade	0.273	0.634	0.0087	0.9910

4.4 Deployment

The full pipeline—inference, token-to-Jianpu mapping, and optional re-engraving of the predicted score—is packaged as an interactive web demo. A user uploads or selects a staff image; the app runs the vit model, renders aligned Jianpu (continuous beams within a beat, octave dots), and can re-engrave the predicted stream for visual comparison. Both the clean baseline and the photo fine-tuned weights are selectable, so the sim-to-real improvement is directly observable on user-supplied photographs.

5 Discussion and Limitations

Why the AR pitch failure was misread. The headline lesson is methodological: an architecture conclusion (“a from-scratch ResNet cannot learn pitch”) was really a statement about the *decoder’s alignment*, not the encoder. Because our four-head design measures pitch separately, and because the CTC variant reuses the identical encoder features, we could falsify the encoder explanation directly. The decoupled-stream contract was what made the negative result interpretable rather than merely discouraging.

Revisiting the alignment bottleneck formally. The same argument can be stated at the level of the decoding objective. Write $\mathbf{H} = (\mathbf{h}_1, \dots, \mathbf{h}_T)$ for the column-pooled ResNet features. The AR decoder factorises the target likelihood autoregressively,

$$P(\mathbf{y} \mid \mathbf{H}) = \prod_u P(y_u \mid y_{<u}, \mathbf{H}),$$

conditioning each step on a cross-attention context $\mathbf{c}_u = \sum_t \alpha_{u,t} \mathbf{h}_t$. Because column pooling makes the features translation-equivariant—discarding the *absolute* vertical position that is pitch on a staff—and the attention mass $\alpha_{u,t}$ concentrates on a narrow temporal window per symbol, the decoder is starved of the global staff-line context needed to infer absolute pitch. CTC instead marginalises over every valid alignment path π under the many-to-one collapse $\mathcal{B}$,

$$P(\mathbf{y} \mid \mathbf{H}) = \sum_{\pi \in \mathcal{B}^{-1}(\mathbf{y})} \prod_t P(\pi_t \mid \mathbf{H}),$$

reading from BiLSTM-contextualised features $\mathbf{h}'_t = \text{BiLSTM}(\mathbf{H})_t$. The marginalisation dissolves the rigid one-column-per-symbol commitment, while the bidirectional context lets each column aggregate the staff-line geometry of its neighbours into the vertical reference frame that column pooling removed—recovering pitch in full, and without any explicit 2-D positional embedding. This is why the *identical* encoder features that look pitch-blind under AR (§4.1) become perfectly legible under CTC.

Pretraining is not a free low-data prior. The sample-efficiency gap (§4.2) cautions against assuming ImageNet-pretrained encoders help most when data is scarce. For a task whose key cue—absolute vertical position—is orthogonal to natural-image semantics, the prior contributes little, and the inductive bias of the decoder (CTC’s alignment flexibility) dominates sample efficiency.

Limitations. (i) *Scope.* We handle only printed, monophonic notation; chords, slurs, multi-voice scores, and handwriting are out of scope, and real-world utility on polyphonic or hand-written music remains untested. (ii) *No external baselines.* Our comparisons are internal (ViT-AR vs ResNet-AR vs CRNN+CTC). We did not run the pipeline OMR baseline (Oemer) or a pretrained Vision-Encoder-Decoder OCR baseline (TrOCR) end-to-end on this task; a head-to-head against those systems is future work. (iii) *Statistical power.* The scaling sweep uses a single seed per cell, so we report trends, not confidence intervals; the ViT-AR data threshold in particular may move with seed and data-loader settings, though the order-of-magnitude gap to CRNN+CTC is stable. (iv) *Sim-to-real ceiling.* Even after fine-tuning, real-photo pitch accuracy is 64.5%—useful but far from the clean 99.8%; the dominant residual is camera-angle vertical mis-registration, which the height-preserving CRNN+CTC encoder is well-positioned to attack but which we have not yet combined with the photo augmentation. (v) *Deployment gap.* The best model (CRNN+CTC) is not yet the deployed one: its CTC decoder needs the type-anchored collapse wired into the inference path, and its recurrent weights require a serialization fix to survive checkpoint reloading. The deployed demo therefore ships the ViT-AR model.

Future work. The natural next step is to bring CRNN+CTC—which is both more accurate and far more sample-efficient—through deployment, then fine-tune *it* with the photo augmentation, since its preserved vertical resolution directly targets the sim-to-real bottleneck. Adding the external Oemer/TrOCR baselines would situate our system against the broader OMR literature, and extending the generator toward polyphony would test how far the four-stream contract scales.

6 Conclusion

We built an end-to-end system that transcribes printed monophonic staff notation directly into Jianpu, trained entirely on a controlled synthetic corpus whose generator emits four decoupled label streams as exact ground truth. The decoupled-stream design turned an architecture comparison into a diagnosis: a pretrained ViT encoder with an autoregressive decoder solves the task ($SER = 0.0029$), a from-scratch ResNet encoder appears to fail at pitch under the same decoder, yet a CTC decoder over the *identical* ResNet features attains a perfect $SER = 0.0000$ —locating the bottleneck in the decoder’s monotonic alignment rather than the encoder. The same CTC model is also far more sample-efficient, reaching usable quality from 5,000 samples where ViT-AR needs 50,000, and a targeted photo-augmentation fine-tune more than halves error on real photographs, with geometric (camera-angle) augmentation the dominant lever. Together these results argue that, for music notation, the decoder’s alignment inductive bias—not the choice of pretrained visual encoder—is what governs whether and how cheaply pitch is learned.

Author Contributions

The project is organised into four loosely-coupled modules connected by a shared batch/token contract, which let the authors work in parallel and integrate through reviewed pull requests.

- **MENG-XI LIN** — *Data*: the synthetic score generator, the music21/verovio renderer, the closed-set four-stream vocabularies, and the multi-encoder data loaders.
- **CHUN-JUI HSU** — *Model*: the Vision-Encoder-Decoder architecture (ViT/ResNet encoders, the multi-head decoder, the CRNN+CTC variant), the per-head losses, and the training loop, including the architecture, sample-efficiency, and photo fine-tune experiments.
- **CHIEN-AN CHOU** — *Post-processing*: the deterministic mapping from the four decoupled token streams to formatted Jianpu output.
- **BEN-LU CHEN** — *Deployment & integration*: the end-to-end inference pipeline, the command-line tool, the interactive web demo, and overall cross-module integration and coordination.

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